POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2014
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Differentiate between Intelligence and Artificial Intelligence. What are the application areas of AI? State in brief. 8

b) You are given two jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug? Solve by production rule system. 7

a) What is state space search? Given any example of a game which happens to be a problem of state space search and justify your answer properly. 8

b) Explain Min-max search. Explain the advantages of Alpha-Beta Pruning over Min-max search? 7

a) Compare depth limited search and iterative deepening search in terms of time and space complexity, completeness and optimality. 8

b) Differentiate propositional and a predicate statement. How does first order predicate logic extend propositional logic to overcome its limitations? 7

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a) What are the problems with hill climbing and how can they be solved? Define plateau 7

b) Represent the following sentences in first order logic: 8

i. Nabin is intelligent than all other students in his class.
ii. Neither Sabin nor Kabin is unhappy.
iii. Same cats are domestic pets
iv. Some turtles are faster than rabbits.

a) Explain how statistical reasoning aids in inference and reasoning in light of Bayes theorem. 7

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b) Differentiate semantic nets and frame based knowledge representation. 8

a) State and explain in brief about EBL-system. What are its advantages 7

b) Explain the different steps involved in the natural language processing (NLP) with suitable block diagram and examples. 8

Write short notes on: ***(Any two)*** 2×5

a) Turjng test

b) Conceptual dependencies

c) Expert System

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